Introduction to data compression, basics of information theory, entropy, …

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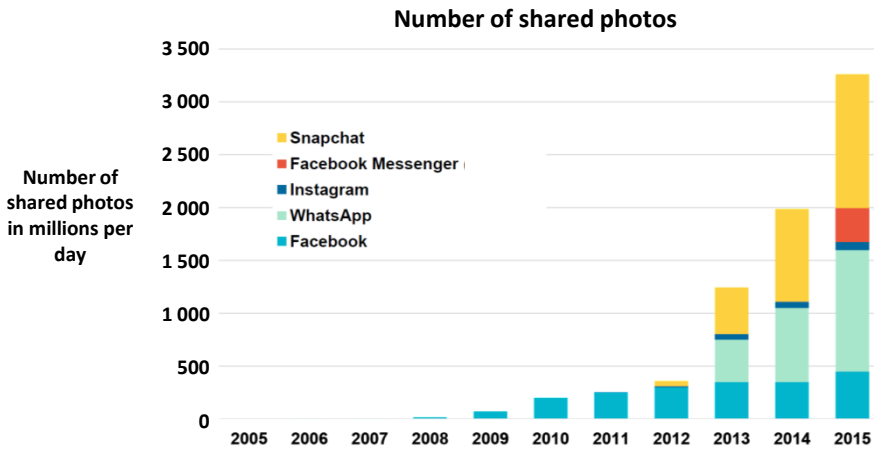
Introduction: Basic motivation

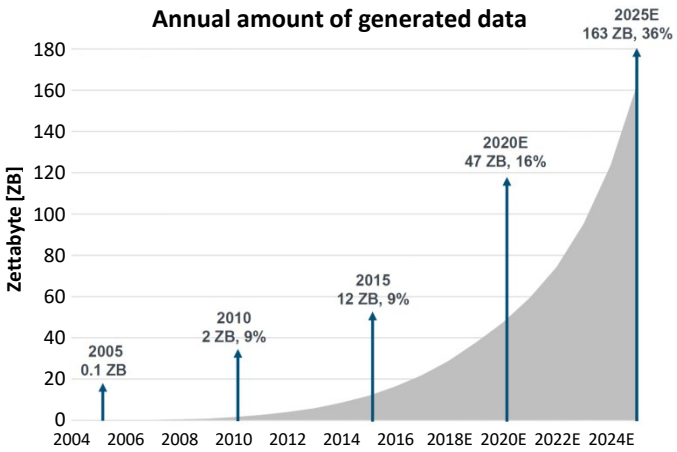
- **Data compression**
  - Art or science of **representing information** in **compact form**
  - Algorithms **reduce** the **number of bits** to represent...
    - ... image, video sequence, music, text, ...
- **Motivation** for **data compression**
  - **More** and **more information** generated in **digital form**
    - Data is to be **transmitted** and **stored efficiently**
      - Explosive development in communication systems, ...
      - ..., Internet, mobile communication and mainly video communication, ...
      - ..., scientific applications, astronomy, medical applications,... etc.
      - **Data compression** important for **all mentioned fields**
  - **Disproportion** in **amount of data** and systems' **capacities**
    - **Need** for **transmission** (bandwidth) and mass **storage** (capacity) grows at least **twice as fast** than storage and transmission **capacities improvement**

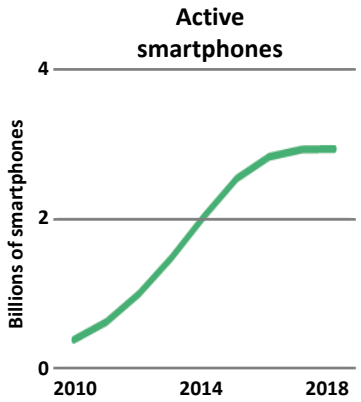
Introduction: Basic motivation

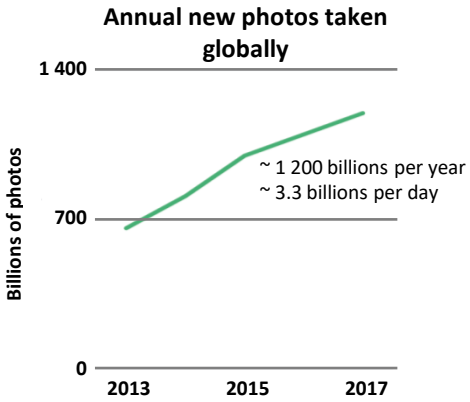
Visual content – majority of generated data

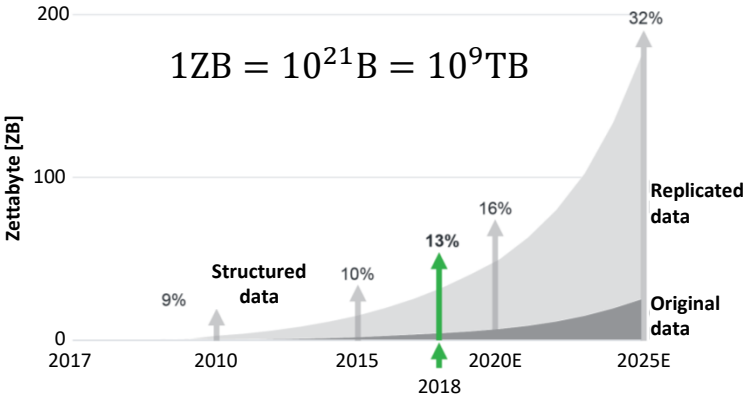
- More than **3 billion photos** shared **every day** via **social nets**
  - Streaming **videos**, sharing **images**, social nets, ...









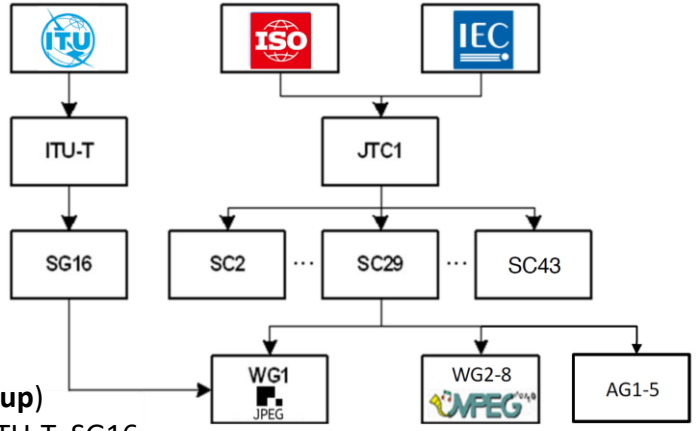


KPCB Internet Trends 2016, 2017, 2018, 2019. (www.bondcap.com)

Introduction: Basic motivation

Standardization in **image** and **video coding**

- Major **standardization committees** in **this field**
  - **Joint Photographic Experts Group (JPEG)**
  - **Moving Picture Experts Group (MPEG)**



JPEG (Joint Photographic Experts Group)
ISO/IEC JTC 1/SC 29/**WG 1** and ITU-T SG16

MPEG (Moving Picture Experts Group)
ISO/IEC JTC 1/SC 29/**WG 2-8** (WG11) and ITU-T SG16

- **Both groups awarded multiple times** (e.g. Emmy, ...)
  - **Television Academy** (Technology and Engineering Emmy Awards)

Emmy Award
MPEG and JPEG
(1995-96)

Emmy Award
JPEG „JPEG 1992“
(2019)

71st Engineering Emmy Award
for **JPEG 1**
(23. 10. 2019
Los Angeles, USA)

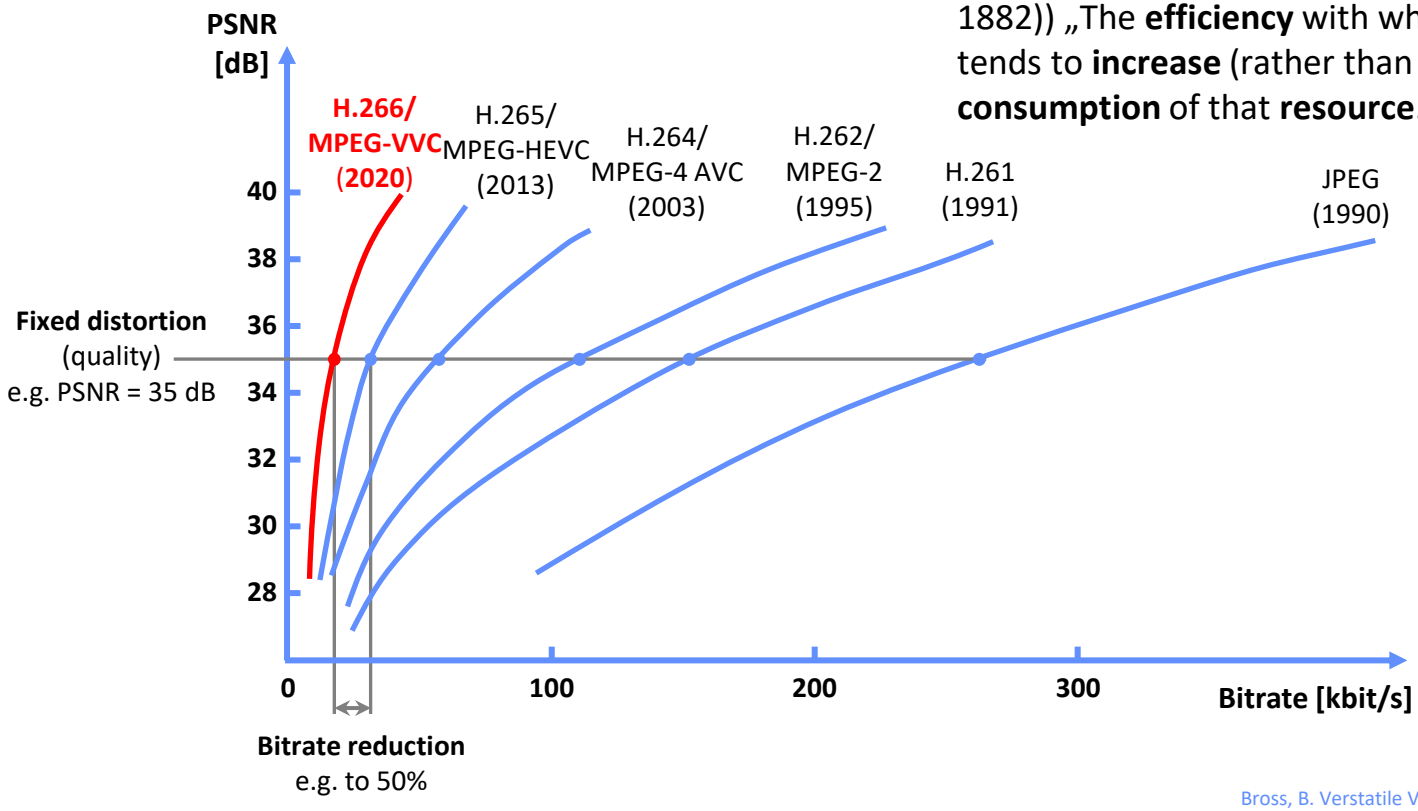
77th Engineering Emmy Award
for **JPEG XS**
(14. 10. 2025
Los Angeles, USA)

Introduction: Basic motivation

Coding efficiency in standards' generations

Defining requirements for coding efficiency

- **Using RD** (rate/distortion) **curves - bitrate vs. distortion** (quality)
- **Is it still necessary** to **improve coding efficiency?**
  - **Jevons Paradox** (William Stanley Jevons (1835-1882)) „The **efficiency** with which a **resource** is **used** tends to **increase** (rather than decrease) the **rate** of **consumption** of that **resource**."

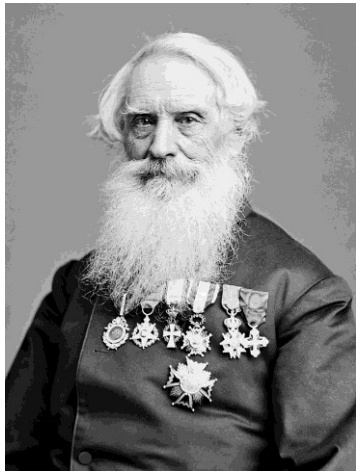


Bross, B. Verstatile Video Coding (VVC) on the final stretch, 2019.

Introduction: History

Morse code

- **Samuel Morse** (1791 – 1872)
  - American sculptor, painter, inventor, ...
- **Telegraph** transmission of „dots“ and „dashes“
- **Different frequency** of letter occurrence
- **Average time reduction** to send message
  - **Shorter sequences** for **frequent letters**, e.g. *e* (·), *a* (·–), **longer sequences** for **less frequent letters**, e.g. *q* (– – · –), *j* (· – – –)
  - Similar **idea** is used in **Huffman coding**



Braille coding

- **Louis Braille** (1809 – 1852)
  - French teacher and inventor, visually impaired since child
- Text represented using 2×3 arrays of dots
  - Some of 2^6 combinations **represent frequent words**

Introduction: History

- What is **exploited** in **signal compression**...?
  - Mainly **statistical structure** of signals
    - **Morse** code and **Braille** code, ...

- Other **exploited characteristics** and examples
  - (**I**) Known signal **source model**
    - Typically in voice coding **mechanics** of **speech production known**
      - **Information** can be **transmitted** on **conformation** of **voice box**
      - **Voice coder** (*vocoder*), Bell Laboratories 1936, major attraction at New York World's Fair 1939
  - (**II**) Known **characteristics** of data **user** (consumer)
    - In **multimedia** systems **users** mostly **human observers/listeners**
    - **Limited perceptual abilities** → **discarding irrelevant information**
      - **Discarding** frequencies, which **dogs hear** but **humans not**, no reason for that...
      - **Discarding** details in colors, which **cannot be seen**, no reason for that...

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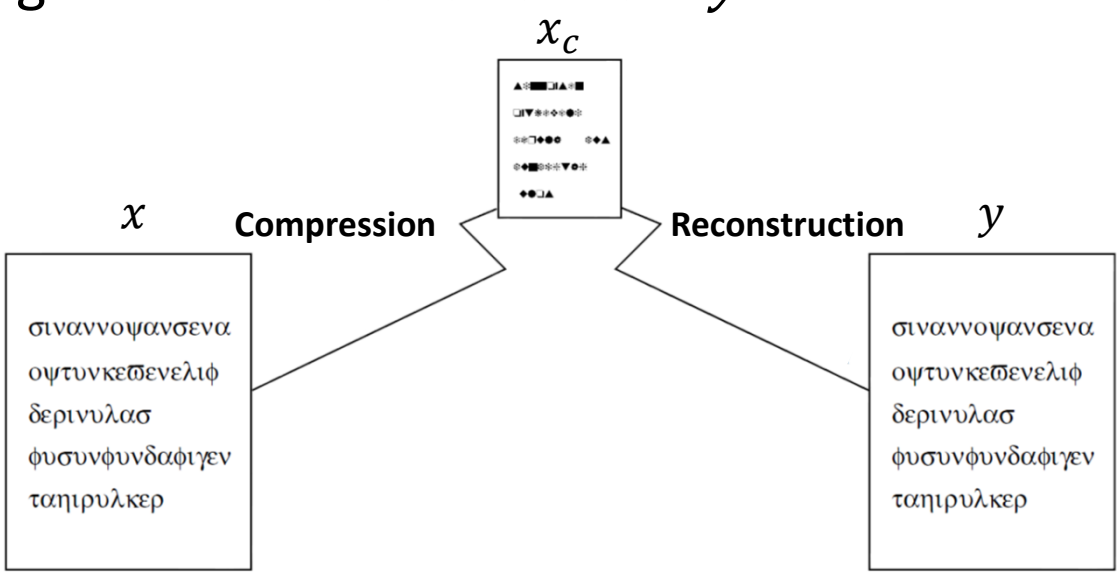
Introduction: Compression

- Basic **elements** and **description**
  - **Compression** algorithm – takes **input** x and generates **representation** x_c requiring **fewer bits**
  - **Reconstruction** algorithm – takes compressed **representation** x_c and generates **reconstruction** y



  - (Mathematically) **lossless** compression $x = y$
  - (Mathematically) **lossy** compression $x \neq y$

Sayood, K.: Introduction to Data Compression, Morgan Kaufmann Publishers, 2006.

Introduction: Compression

- **Lossless** and **lossy** compression
  - **Basic classification**
    - Basic categories can be further specified ...
  - **Lossless** compression
    - **Original** data x can be **recovered exactly** from compressed x_c
    - **Cannot tolerate any difference** between x and y
    - Example is **text compression**, **compression** in **medical imaging**, ...
  - **Lossy** compression
    - **Original** data x **cannot be recovered/reconstructed exactly** from x_c
    - **Difference** between x and y **can be tolerated**
    - Example is **speech/audio**, **multimedia image compression**, ...
      - Necessary to **consider requirements** for **quality** of **reconstruction**
      - **Note** – **lossy** compression can be **perceptually lossless**

Introduction: Compression

Measures of compression performance

- Different **ways** of **measuring** ...
  - Algorithm complexity, memory requirements, speed, amount of compression, how close reconstruction resembles original,...
- **Compression ratio**
  - **Ratio** of **number** of **bits** required to represent data **before** and **after compression**
    - **Example** – Image 256×256 pixels, storage required $65\,536$ byte, after compression $16\,384$ byte, **compression ratio** $4{:}\,1$, can be represented as **reduction** in **amount** of **data** 75%
  - Another way using **average number** of **bits** required to **represent single sample** (*compression rate*)
    - **Example** – original image 8 bpp, after compression 2 bpp (*bits per pixel*)
- **Evaluation of distortion/fidelity/quality** of reconstruction
  - **Lossy compression** – **measuring difference** original x vs. reconstruction y
  - Purely **mathematical** approaches and **perceptual** approaches

Introduction: Modeling and coding

Modeling and coding

- **Requirements** on **compression scheme** (**lossy** X **lossless**)
- Selected **compression scheme** depends on **various factors**
  - For instance, scheme for **text compression** will **not perform** well in **image compression**

- **Development** of data **compression** has **two phases…**
  - (**i**) **Modeling** – extracts **information about redundancy** in data and **describes redundancy** in form of **model**
  - (**ii**) **Coding** – definition of **model** and description **how data differ from this model** as **residual** are **encoded**

Introduction: Modeling and coding

Modeling and coding

Example (1)

- Sequence of numbers $x_n = \{x_1, x_2, x_3, \dots\}$

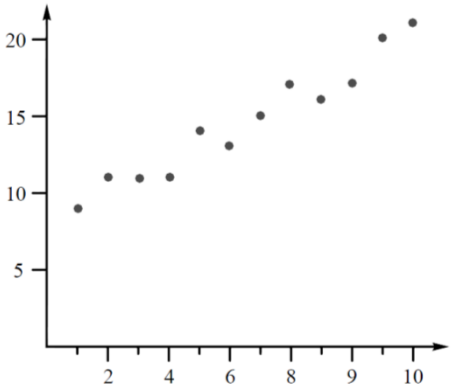
9	11	11	11	14	13	15	17	16	17	20	21

- Using **binary representation** we need **5 bits per sample**
- Exploiting **structure** in **data** (**model**)

$$\hat{x}_n = n + 8 \qquad n = 1,2,\dots$$

- **Difference** (*residual*) is given by sequence

$$e_n = x_n - \hat{x}_n = \{0,1,0,-1,1,-1,0,1,-1,-1,1,1\}$$



  – Residual consists of **only 3 numbers** $\{-1,0,1\}$
  – **Assigning code** $-1 \to 00,\ 0 \to 01,\ 1 \to 10$

- Exploiting **predictive coding** only **2 bits represent** each element in **residual sequence**
  – Encoding **exact** for **lossless** compression or **approximate** for **lossy** compression

Sayood, K.: Introduction to Data Compression, Morgan Kaufmann Publishers, 2006.

Introduction: Modeling and coding

Modeling and coding

Example (2)

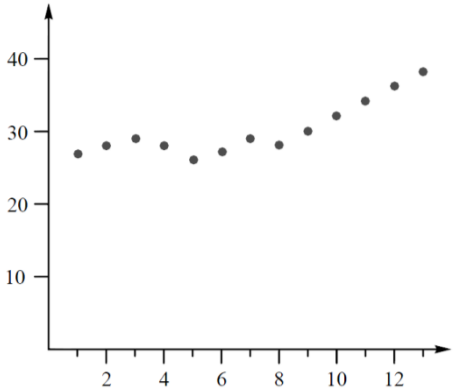
- Sequence of numbers $x_n = \{x_1, x_2, x_3, \dots\}$

27	28	29	28	26	27	29	28	30	32	34	36	38

- Does **not follow** previous simple law, but **each value close to previous**
- Transmission of **first value** and in place of **subsequent values** only **difference** is sent

27	1	1	−1	−2	1	2	−1	2	2	2	2	2

- Decoder **adds received value** to previously decoded value
  - It is necessary to send also the **first value** ...
- Technique uses past values to **predict** current in **predictive coding**
  - There are lossless and lossy schemes



Sayood, K.: Introduction to Data Compression, Morgan Kaufmann Publishers, 2006.

Introduction: Modeling and coding

Modeling and coding

Example (3)

- Sequence of symbols

  a~~b~~barrayaran~~b~~array~~b~~ran~~b~~far~~b~~far~~b~~far~~b~~away

  – Suppose it is **typical** of **all sequences generated** by **source**

  – **Sequence** made up of **8 different symbols**

  – In order to **represent** them we need **3 bits per symbol**

- Using **short codeword** for **symbols that occur often**

  – Using **dictionary compression scheme**

  – Using **106 bits** to encode **entire sequence**

  – **41 symbols in sequence**, works out approximately **2.59 bits per symbol**

- **Compression ratio** is **1.16:1**

- Using **statistical redundancy**

  – Used in **Huffman** and **arithmetic coding**

  – **Dictionary compression scheme**

a	1
~~*b*~~	001
b	01100
f	0100
n	0111
r	000
w	01101
y	0101

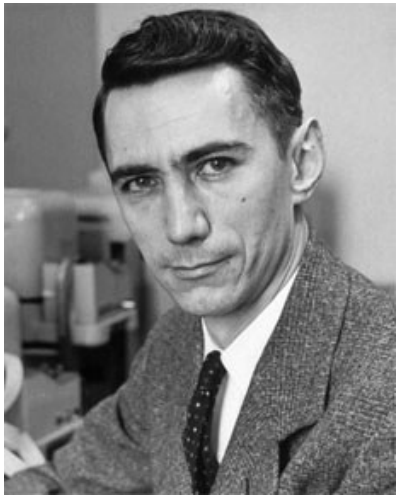
Sayood, K.: Introduction to Data Compression, Morgan Kaufmann Publishers, 2006.

Mathematical preliminaries for lossless compression: Introduction to information theory

- **Mathematical treatment** of **data compression**
  - In this **introductory** course **reduced** to **minimum**
    - Only **necessary theory** for **understanding compression methods**
  - **Information theory** used in **development** of data compression
- **Information theory**
    - **Scientific field** – measurement, transmission, coding, storage and subsequent **processing** of **information** from **quantitative point** of **view**
  - **Claude Elwood Shannon** (1916 – 2001)
    - American mathematician, electrical engineer, „father of information theory“
  - Defined quantity **self-information** (of event A)

$$i(A) = \log_b \frac{1}{P(A)} = -\log_b P(A)$$

    - Event A is a set of **outcomes** of **random experiment**
    - **Probability** $P(A)$ that **event** A will **occur**



Mathematical preliminaries for lossless compression: Introduction to information theory

Self-information

$$i(A) = \log_b \frac{1}{P(A)} = -\log_b P(A)$$

- Event A as **outcome** of **random experiment**
- **Probability** $P(A)$ that **event** A **will occur**
  - Base of the logarithm b is not specified yet
  - $\log(1) = 0$ and $-\log(x)$ **increases** as x **decreases** from one to zero
  - **Probability** of event **low** → **high** amount of **self-information**
    - **Barking dog** during burglary (high-probability) → **not much information**
    - **Silent dog** during burglary (low-probability) → **lot of information**
    - **Totally random string** of **letters** → **more information** in mathematical sense
    - **Structured texts** → **less information** in mathematical sense
- **Unit of information** depends on **base** of **logarithm** b

  $b = 2$ – unit is **bit** [**bit**, Shannon – *binary digit*]

  $b = e$ – unit is **nat** [**nat**, nit, nepit – *natural unit of information*]

  $b = 10$ – unit is **Hartley** [**Hart**, ban, dit – *decimal digit*]

Mathematical preliminaries for lossless compression: Introduction to information theory

Self-information

- Associated with occurrence of **two independent events** A, B

$$i(AB) = \log_b \frac{1}{P(AB)}$$

  - **Independent events** A and B

$$P(AB) = P(A)P(B)$$

$$i(AB) = \log_b \frac{1}{P(A)P(B)}$$

$$= \log_b \frac{1}{P(A)} + \log_b \frac{1}{P(B)} = i(A) + i(B)$$

- **Example**
  - **Flipping** a **coin**, events H (*heads*) and T (*tails*)
    - For **fair coin** $P(H) = P(T) = 1/2$
      - Self-information $i(H) = i(T) = 1$ bit
    - For **unfair coin**, e.g. $P(H) = 1/8$ and $P(T) = 7/8$
      - Self-information $i(H) = 3$ bit, $i(T) = 0.193$ bit
      - **Mathematically** occurrence of H **conveys more information** than occurrence of T

Mathematical preliminaries for lossless compression: Introduction to information theory

Self-information and entropy

- For **set of independent events** A_i, which are sets of **outcomes** of **experiment** $\mathcal{S}$, such that for space S applies

$$\cup A_i = S$$

- Then **average self-information** associated with **random experiment** is given by

$$H = \sum P(A_i) i(A_i) = -\sum P(A_i) \log_b P(A_i)$$

  - Quantity is called **entropy** (associated with experiment)
  - **Shannon** showed – if **experiment** is **source that puts out symbols** A_i from set $\mathcal{A}$ then **entropy** is measure of **average number** of **binary symbols** needed to **code source output**
    - Defines limit of **lossy compression** performance

Mathematical preliminaries for lossless compression: Introduction to information theory

Source entropy

- Set of symbols $\mathcal{A}$ is **source alphabet**, for general **source** $\mathcal{S}$ with **alphabet** $\mathcal{A} = \{1,2,\dots,m\}$ that **generates sequence** $\{X_1, X_2, \dots\}$ the **entropy** is

$$H(\mathcal{S}) = \lim_{n\to\infty} \frac{1}{n} G_n$$

$$G_n = \\ = -\sum_{i_1=1}^{m}\sum_{i_2=1}^{m}\cdots\sum_{i_n=1}^{m} P(X_1 = i_1, X_2 = i_2, \dots, X_n = i_n) \log P(X_1 = i_1, X_2 = i_2, \dots, X_n = i_n)$$

and $\{X_1, X_2, \dots, X_n\}$ is **sequence** of length n from source.

- For **independent** and **identically distributed** (iid) sequence elements

$$G_n = -n \sum_{i_1=1}^{m} P(X_1 = i_1) \log P(X_1 = i_1)$$

- **First order entropy** of source

$$H(S) = -\sum_{i_1=1}^{m} P(X_1 = i_1) \log P(X_1 = i_1) = -\sum p_i \log p_i$$

Mathematical preliminaries for lossless compression: Introduction to information theory

Source entropy

- Is **not possible** to **know exactly** but **has to be estimated**
- **Example**
  - Sequence of numbers $\{1, 2, 3, 2, 3, 4, 5, 4, 5, 6, 7, 8, 9, 8, 9, 10\}$
    - Assumption – **frequency** of **occurrence** – **number** of **times it appears**
    - **Probability** of **occurrence** can be **estimated**

$$P(1) = P(6) = P(7) = P(10) = \frac{1}{16}$$

$$P(2) = P(3) = P(4) = P(5) = P(8) = P(9) = \frac{2}{16}$$

    - For ***iid* sequence** entropy equals to first-order entropy

$$H = -\sum_{i=1}^{10} P(i) \log_2 P(i)$$

    - **Source entropy**

$$H = -4\frac{1}{16}\log_2\frac{1}{16} - 6\frac{1}{8}\log_2\frac{1}{8} = \frac{1}{4}4 + \frac{3}{4}3 = \frac{13}{4} = 3.25 \text{ bit}$$

  - **Best compression scheme** can achieve **only 3.25 bits per sample**

Source entropy

Example

- Sequence of numbers $\{1, 2, 3, 2, 3, 4, 5, 4, 5, 6, 7, 8, 9, 8, 9, 10\}$
  - Assuming **sample-to-sample correlation** and **removing** the **correlation**
  - **Residual** sequence $\{1, 1, 1, -1, 1, 1, 1, -1, 1, 1, 1, 1, 1, -1, 1, 1\}$
  - Estimating **probabilities** of **symbols** based on their **occurrence**

$$P(1) = \frac{13}{16} \qquad P(-1) = \frac{3}{16}$$

- **Entropy is only 0.7 bits per symbol**
  - Knowing **residual sequence** is **not enough for reconstruction**
  - Receiver **must also know the process** used for its generation
- **Model** for sequence
  - **Static model** considered

$$x_n = x_{n-1} + r_n$$

  - Also **adaptive models exist** – **parameters change** or **adapt** to **changing data characteristics**

Mathematical preliminaries for lossless compression: Introduction to information theory

Source entropy

- Knowing something **about structure (model)** ...
  - Can help „**reducing entropy**“
    - **Entropy** of the source **remains the same** – amount of information generated
    - **Reducing** only **estimate** of **entropy**
    - **Learning** about **data** – **estimating source entropy** – larger blocks better
- **Example**
  - Contrived sequence $\{1, 2, 1, 2, 3, 3, 3, 3, 1, 2, 3, 3, 3, 3, 1, 2, 3, 3, 1, 2\}$
    - Obvious **structure** in data
    - **Considering one symbol** at **time** – structure **difficult** to **extract**
      $P(1) = P(2) = 1/4$, $P(3) = 1/2$, $H = 1.5$ bits/symbol, 20 symbols → **30 bits in total**
    - **Considering blocks** of **two symbols**
      $P(1, 2) = 1/2$, $P(3, 3) = 1/2$, $H = 1$ bit/symbol, 10 symbols → **10 bits in total**
- Aiming to **accurate model** – remove limitations
  - Allows **lower entropy estimate** and **increase compression performance**

Mathematical preliminaries for lossless compression: Models

Models

- **Entropy estimation** and **improved compression performance**
  - **Mathematical model** for efficient **mathematical manipulation** with data
  - **Better model** allows development of satisfactory **compression technique**
- **(I) Physical models**
  - Use knowledge about **physics** of **data generation**
    - Generally **models difficult** to **develop** – **complicated physics** of **data generation**
      - In speech-related applications e.g. **physics** of **speech production** used in the model
      - For complicated physics – model based on **empirical observation** of **data statistics**
- **(II) Probability models**
  - **Simplest** (ignorance) model – **independent** and **same probability** letters
  - **Model** (probability) with **probabilities** of alphabet letters – **independent**
    - For source with alphabet $\mathcal{A} = \{a_1, a_2, \dots, a_M\}$ **probability model** $\mathcal{P} = \{P(a_1), P(a_2), \dots, P(a_M)\}$ → entropy can be estimated
  - Model with **dependence** of **elements**
    - Necessary find **way** to **describe the dependence** of elements in data sequence

Mathematical preliminaries for lossless compression: Models

Models

- **Markov models**
- **Andrey Andreyevich Markov** (1856 – 1922)
  - Russian mathematician, stochastic processes, …
- Using **discrete time Markov chain**
  - ***k*th-order Markov model** for sequence of observations $\{x_n\}$

$$P(x_n|x_{n-1},\dots,x_{n-k}) = P(x_n|x_{n-1},\dots,x_{n-k},\dots)$$

    - Knowledge of ***k* symbols** is **equivalent** to **knowledge** of **entire past history** of **process**
    - **Values taken** by set $\{x_{n-1},\dots,x_{n-k}\}$ are **states** of **process**, for source alphabet l the number of states is l^k

  - **First-order Markov model** – commonly used

$$P(x_n|x_{n-1}) = P(x_n|x_{n-1},x_{n-2},x_{n-3},\dots)$$

  - Indicate **dependence between samples** – exact **form not described**

Mathematical preliminaries for lossless compression: Models

Models

Markov models

- Often **used** in **compression** of **speech**, **text**, ...

Example (1)

- Assuming **dependence** in **linear manner**
  - Data sequence viewed as **output** of **linear filter driven** by **white noise** ϵ_n

$$x_n = \rho x_{n-1} + \epsilon_n$$

  - Model used commonly in **speech applications**

Example (2)

- Considering **binary image coding** (black and white pixels)
  - **Model** – **probability** of **occurrence** of **white pixel** as the **next observation depends** on whether **current** pixel is **black** or **white**
  - Define **two states** – S_w (current **pixel** is **white**), S_b (current **pixel** is **black**)
  - **Transition** probability $P(w|b)$ and $P(b|w)$
  - **State** probabilities $P(S_w)$ and $P(S_b)$

Mathematical preliminaries for lossless compression: Models

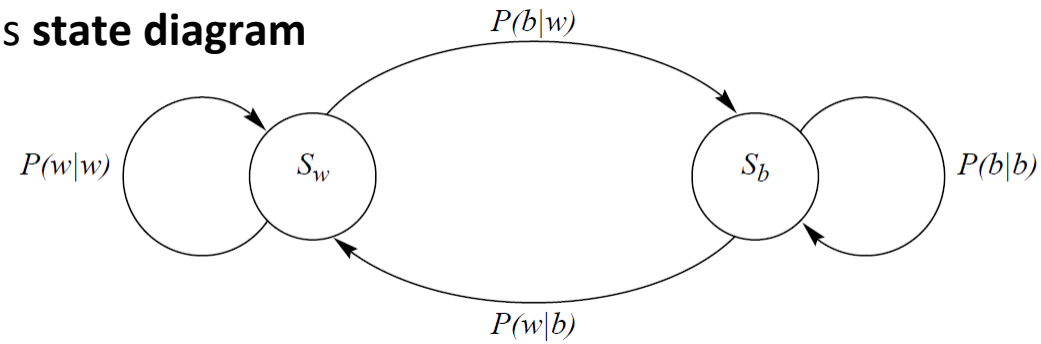
Models

Markov models

Example (2)

- Considering **binary image coding** (black and white pixels)
  - **Markov model** represented as **state diagram**
  - **Entropy** of **finite state** process

$$H = \sum_{i=1}^{M} P(S_i) H(S_i)$$



  - **Average value** of **entropy** at **each state**
  - For our **example** of **binary image**

$$H(S_w) = -P(b|w)\log P(b|w) - P(w|w)\log P(w|w)$$

  - Probability $P(w|w) = 1 - P(b|w)$
  - $H(S_b)$ can be **calculated** in **similar manner**

$$H(S_b) = -P(w|b)\log P(w|b) - P(b|b)\log P(b|b)$$

  - Probability $P(b|b) = 1 - P(w|b)$

Sayood, K.: Introduction to Data Compression, Morgan Kaufmann Publishers, 2006.

Mathematical preliminaries for lossless compression: Models

Models

Markov models

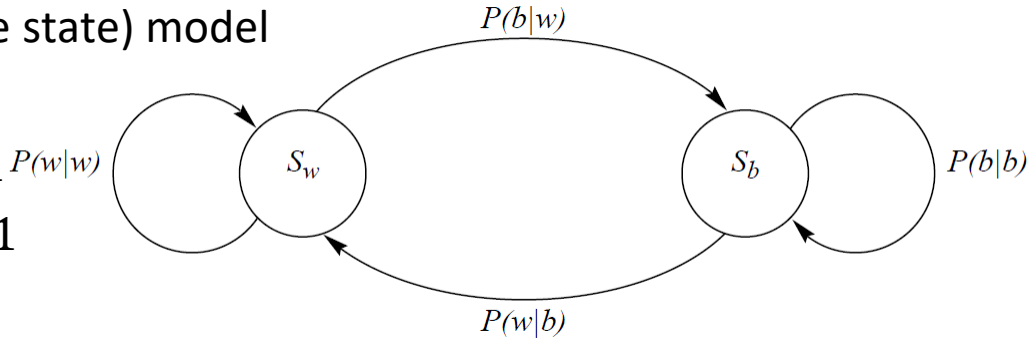
Example (2)

- Considering **binary image coding** (black and white pixels)
  - **Probability** and **Markov** (finite state) model
  - **Definition** of probabilities

$P(S_w) = 30/31, P(S_b) = 1/31$

$P(w|w) = 0.99, P(b|w) = 0.01$

$P(b|b) = 0.70, P(w|b) = 0.30$



  - Using **probability model** and *iid* assumption

$$H = -(30/31)\log(30/31) - (1/31)\log(1/31) = 0.206\text{bit}$$

  - Using **Markov model**

$$H(S_b) = -0.30\log 0.30 - 0.70\log 0.70 = 0.881 \text{ bit}$$
$$H(S_w) = -0.01\log 0.01 - 0.99\log 0.99 = 0.081 \text{ bit}$$
$$H = P(S_b)H(S_b) + P(S_w)H(S_w) = 0.107 \text{ bit}$$

- **Entropy** for **Markov model** is **much lower than** for ***iid* assumption**

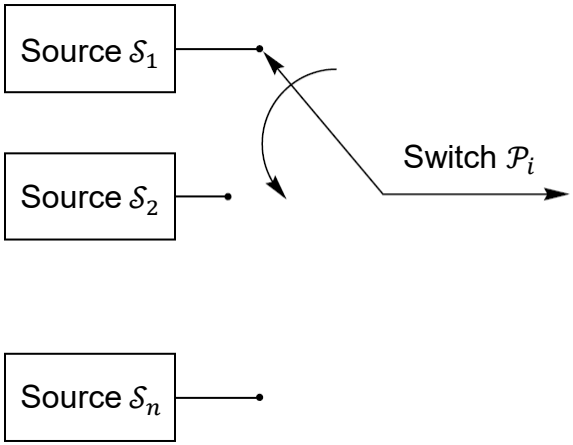
Sayood, K.: Introduction to Data Compression, Morgan Kaufmann Publishers, 2006.

Mathematical preliminaries for lossless compression: Models

Models

Composite source model

- In many applications **not easy** to **use** a **single model** to **describe source**
- **Composite** source – **combination** – only **one source active** at **given time**
  - Represented as **individual sources** $\mathcal{S}_i$, each with **own model** $\mathcal{M}_i$
  - Switch **selects source** $\mathcal{S}_i$ with **probability** $\mathcal{P}_i$
- **Rich model** able to describe **very complicated processes**



Sayood, K.: Introduction to Data Compression, Morgan Kaufmann Publishers, 2006.

Mathematical preliminaries for lossless compression: Coding

Meaning of coding

- Assignment of **binary sequences** to **elements** of **alphabet**
  - **Set** of **binary sequences** is called **code** and **individual members** of the set are **codewords**
  - **Alphabet** is a **collection** of **symbols** called **letters**
- **Example**
  - Basic **English alphabet** has 26 lowercase, 26 uppercase **letters** and variety of punctuation marks (e.g. comma (,) is a letter)
    - ASCII (American Standard Code for Information Interchange, 1960) **code** for **letter** *a* is 1000011, for letter A is 1000001, and for letter (,) is 0011010,...
    - Such code is **fixed-length code**
    - **Bits reduction** to represent different symbols → **variable length code**
      - **Often occurring** symbols – **fewer bits** (shorter codeword)
      - **Rarely occurring** symbols – **more bits** (longer codeword)
    - Typical example **Morse code**

Mathematical preliminaries for lossless compression: Coding

Uniquely decodable codes

- Important is **average number** of **bits per symbol** called **rate**
  - **Designing "good" code** considering **other important points**
  - **Unique decodability**
    - Each **symbol** is assigned **unique codeword**
- **Example**
  - Source alphabet has **4 letters** $\{a_1, a_2, a_3, a_4\}$
    - Related **probabilities** $P(a_1) = 1/2, P(a_2) = 1/4, P(a_3) = P(a_4) = 1/8$, source **entropy** is 1.75 bit/symbol
    - **Different codes** for four-letter alphabet

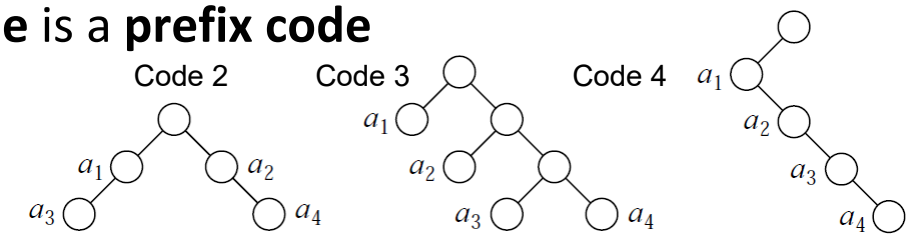
Letters	Probability	Code 1	Code 2	Code 3	Code 4
a_1	0.5	0	0	0	0
a_2	0.25	0	1	10	01
a_3	0.125	1	00	110	011
a_4	0.125	10	11	111	0111
Average length		1.125	1.25	1.75	1.875

- Code 1 and 2 – **not uniquely decodable**
- Code 3 and 4 – **uniquely decodable**
- Code 3 – **instantaneous**, no need to wait for next codeword

Mathematical preliminaries for lossless compression: Coding

Prefix codes

- **No need** to **check unique decodability** of **prefix codes**
  - No need to use **dangling suffix** to **separate particular codewords**
  - **Prefix codes** – **no codeword** is **prefix** to **another codeword** (e.g. **Huffman code**)
- **Code** can be **represented** with **rooted binary tree**
  - Way to **check** if **code** is a **prefix code**



Letters	Probability	Code 1	Code 2	Code 3	Code 4
a_1	0.5	0	0	0	0
a_2	0.25	0	1	10	01
a_3	0.125	1	00	110	011
a_4	0.125	10	11	111	0111
Average length		1.125	1.25	1.75	1.875

- **Prefix code** has only **external nodes** or **leaves**
  - Code 4 is **not** a **prefix code**

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Mathematical preliminaries for lossless compression: Coding

Kraft-McMillan inequality

- **Necessary conditions** on **codeword lengths** – to be **uniquely decodable** and **prefix** codes
- Considering **binary uniquely decodable code** $\mathcal{C}$ with N **codewords** with **lengths** $\{l_1, l_2, \dots, l_N\}$. Then **inequality applies**

$$K(\mathcal{C}) = \sum_{i=1}^{N} 2^{-l_i} \leq 1.$$

  – **Necessary condition** for **uniquely decodable** codes
  – **Codeword lengths** of **uniquely decodable** codes **must satisfy** the **inequality**

- **Given** set of **integers** $\{l_1, l_2, \dots, l_N\}$ that **satisfy** the **inequality**

$$\sum_{i=1}^{N} 2^{-l_i} \leq 1$$

  we can **always find** a **prefix code** with **codeword lengths** $\{l_1, l_2, \dots, l_N\}$.
  – **Restricting** to use of **prefix codes** will **not increase the average length** in respect to **nonprefix uniquely decodable codes**

Mathematical preliminaries for lossless compression: Algorithmic information theory

Algorithmic information theory

- **Information theory** have some **theoretical difficulties**
  - **Example** – **set** of **documents** – **entire set** as **single long string**
    - **Models** for this **data** can be developed
    - **Probabilities** based on **relative frequency approach** can be calculated
    - **Estimate** of **entropy** and **amount** of **available compression** can be obtained
  - **Given string is fixed** – **nothing probabilistic** about it
    - No **abstract source** generating **different sets** of **documents**
  - Such **entropy estimate** has its **limitations** – but **still usefull** approximation
- **Kolmogorov complexity** provides **solution**
  - **Andrey Nikolaevich Kolmogorov** (1903 – 1987)
    - Russian mathematician
  - **Kolmogorov complexity** $K(x)$ of sequence x is **size of program** needed to generate x
    - Sequence of the **same letters** (print in **loop**)
    - **Random** sequence (program **contains** the **sequence** itself)

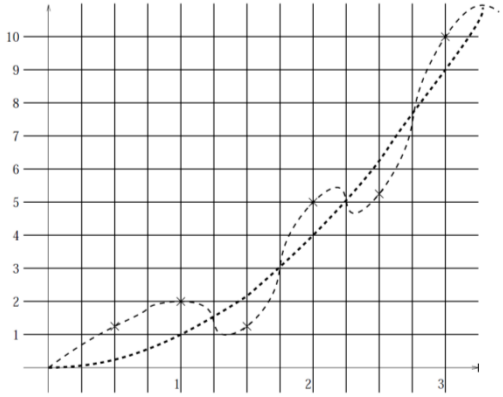
Mathematical preliminaries for lossless compression: Minimum description length principle

Minimum description length principle

- Practical offshoot of **Kolmogorov complexity**
  - MDL – *Minimum Description Length*
  - **Structure** in **sequence** x is **characterized** by **set** of **models** $\mathcal{M}$ including **models** M_j
    - **Number** of **bits** D_{M_j} required to **describe model** M_j
    - For example **set** of **models** $\mathcal{M}$, where **description** of **each model** M_j include number of **coefficients**
    - **Number** of **bits** $R_{M_j}(x)$ required to **represent sequence** x using model M_j
  - **MDL** is **given** by

$$\min_j (D_{M_j} + R_{M_j}(x))$$

  - **Example** – set of **models** $\mathcal{M}$ is $\boldsymbol{k}$**th order polynomials set**
    - **Higher** order - **better modeling** → **more coefficients**



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Thank you for your attention!

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